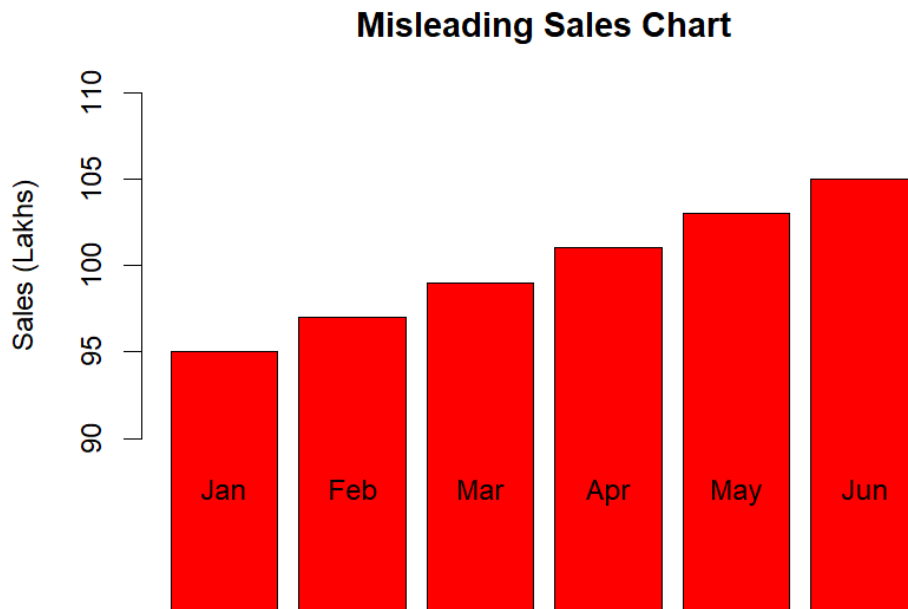


## ACTIVITY-6

### Case Analysis – Avoiding Misleading Proportions

#### QUESTION-1 Truncated Y-Axis in Sales Growth

##### Misleading Graph:



#### 1. Explain why the chart is misleading.

The chart is misleading because the Y-axis starts at 90 instead of 0. This exaggerates the difference between January (95 lakhs) and June (105 lakhs). The actual increase is only ₹10 lakhs (about 10.5%), but visually it appears much larger, making June sales seem almost twice as high as January.

#### 2. Identify the visualization principle that has been violated.

**Principle Violated:** Honest representation of data (Integrity in visualization).

Bar charts should normally start the Y-axis at zero so that the bar lengths accurately represent the values.

### 3. Recommend an improved visualization method.

- Use a bar chart with the Y-axis starting at 0.
- Alternatively, use a line chart to show monthly sales trends without exaggerating differences.

### 4. Describe how this misleading chart could influence business decisions.

The misleading chart may cause managers to believe sales have increased dramatically. This could result in:

- Overestimating business growth.
- Increasing production unnecessarily.
- Making poor budgeting decisions.
- Misleading investors and stakeholders.

### Question 2: Pie Chart with Incorrect Proportions

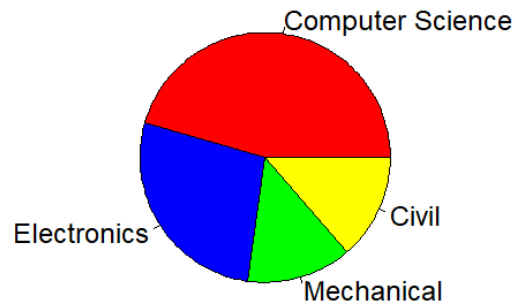
#### Given Data

| Department       | Students |
|------------------|----------|
| Computer Science | 420      |
| Electronics      | 280      |
| Mechanical       | 200      |
| Civil            | 100      |

#### Total Students

$$420 + 280 + 200 + 100 = \mathbf{1000}$$

### Incorrect Pie Chart



#### 1. Calculate the correct percentage for each department.

| Department       | Students | Percentage |
|------------------|----------|------------|
| Computer Science | 420      | 42%        |
| Electronics      | 280      | 28%        |
| Mechanical       | 200      | 20%        |
| Civil            | 100      | 10%        |

#### 2. Explain why the displayed chart is misleading.

The displayed pie chart uses incorrect slice sizes (50%, 30%, 15%, 15%), which do not match the actual data. This gives a false impression of student enrollment in each department.

### 3. Suggest an alternative chart that would improve comparison accuracy.

**A bar chart is a better choice because:**

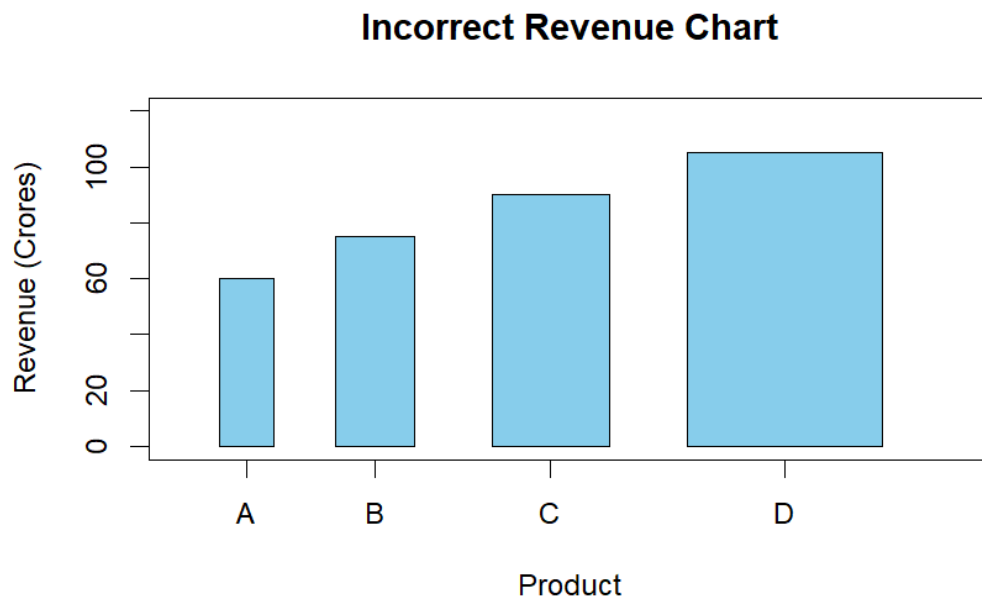
- It allows easier comparison between departments.
- Differences in values are clearer than in a pie chart.
- It reduces the chance of proportion errors.

### 4. Discuss how such errors could affect institutional planning.

**Incorrect proportions may lead to:**

- Improper allocation of faculty.
- Wrong classroom planning.
- Incorrect budget distribution.
- Poor admission planning.

### Question 3: Unequal Bar Widths in Revenue Comparison



#### 1. Identify the graphical error.

The designer used unequal bar widths.

Only the height should represent revenue, while all bars must have equal width.

## **2. Explain how unequal bar widths distort perception.**

Wider bars occupy more area, making Product D appear much larger than Product A, even though the actual revenue difference is only:

$$105 - 60 = ₹45 \text{ crores}$$

The larger area tricks viewers into believing Product D generates several times more revenue.

## **3. Redesign the visualization by specifying the correct design principles.**

**Correct design should include:**

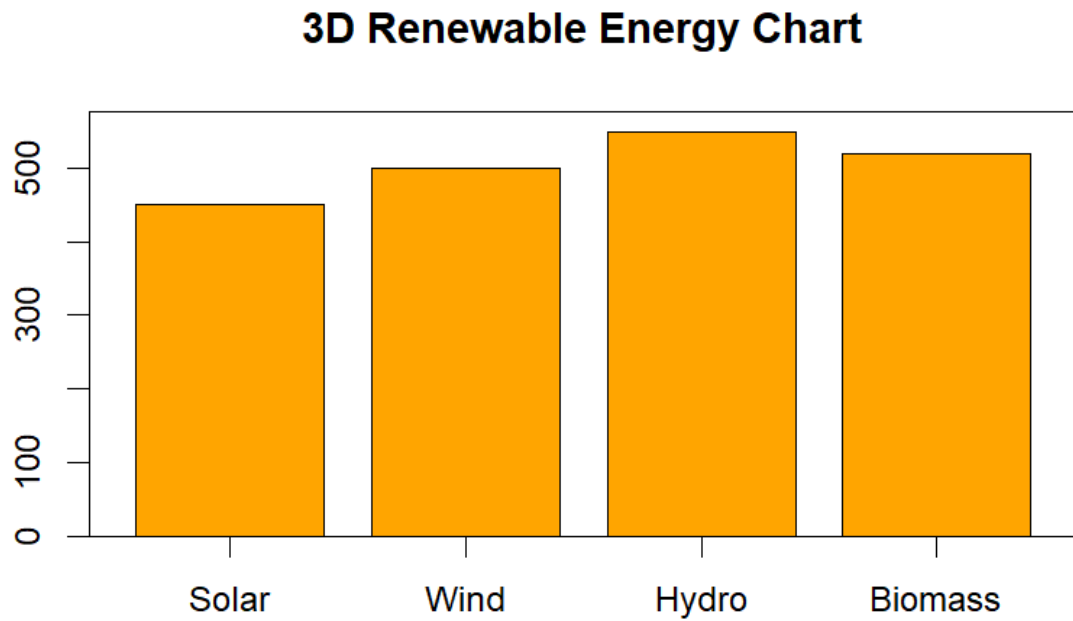
- Equal-width bars.
- Heights proportional to revenue.
- Y-axis starting at zero.
- Uniform spacing between bars.
- Clear labels and title.

## **4. Discuss the possible consequences if investors rely on this chart.**

**Investors may:**

- Overestimate Product D's performance.
- Invest based on misleading information.
- Lose confidence if actual financial reports differ.
- Make poor investment decisions.

#### Question 4: 3D Column Chart Distortion



##### 1. Explain why 3D charts can create misleading proportions.

3D charts introduce perspective effects that make some bars appear taller or larger than they actually are. Viewers cannot accurately compare values because depth creates visual distortion.

##### 2. Compare the perceived difference with the actual difference.

Actual values:

- Wind = 500 GWh
- Hydro = 550 GWh

Actual Difference:

$$550 - 500 = 50 \text{ GWh}$$

Percentage Difference:

$$\frac{50}{500} \times 100 = 10\%$$

Actual increase = 10%

However, due to the 3D effect, Hydro appears approximately 40% larger, greatly exaggerating the real difference.

### 3. Recommend a better visualization.

Use a:

- 2D bar chart
- Horizontal bar chart

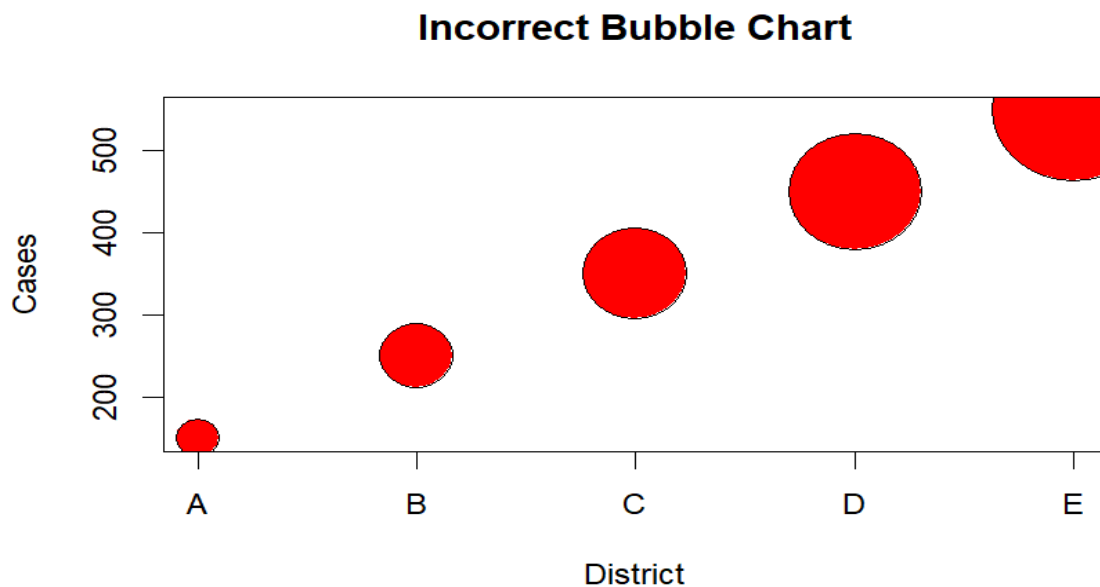
These accurately display differences without distortion.

### 4. Discuss the importance of avoiding visual distortion in sustainability reporting.

Accurate visualizations help:

- Build public trust.
- Support transparent reporting.
- Enable better policy decisions.
- Prevent incorrect conclusions about renewable energy performance.

#### Question 5: Bubble Chart with Incorrect Scaling



### **1. Explain why scaling the radius instead of the area is misleading.**

If the radius is directly proportional to the number of cases, then the area increases much faster because:

$$\text{Area} = \pi r^2$$

As a result, District E appears much larger than it should, exaggerating the difference between districts.

### **2. Describe the correct method for proportional bubble sizing.**

The area of each bubble should be proportional to the data value.

Therefore,

$$r \propto \sqrt{\text{Value}}$$

This ensures that viewers correctly perceive the relative sizes.

### **3. Suggest an alternative visualization suitable for this dataset.**

A bar chart is more suitable because:

- It provides accurate comparisons.
- It is easy to understand.
- It avoids area perception errors.

### **4. Discuss the potential impact of this visualization on public health resource allocation.**

**A misleading bubble chart may:**

- Allocate too many medical resources to one district.
- Underestimate the needs of smaller districts.
- Affect vaccination and medicine distribution.
- Lead to poor public health planning.